

Active Inference for Joint Port Selection and Pilot Allocation in Fluid Antenna Systems Under Partial CSI

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Abstract—Fluid antenna systems (FAS) obtain spatial degrees of freedom by activating a few radiating ports from a dense grid of candidates, but deciding which ports to activate presupposes channel knowledge at ports that are never measured. Prior port-selection work assumes full or near-full channel state information (CSI) and treats the pilot overhead as a fixed cost. We instead formulate port selection and pilot allocation *jointly* as active inference. A spatio-temporal generative model — sub-wavelength spatial correlation together with an order- p autoregressive temporal prior — maintains a Gaussian belief over all N port channels, and the transmitter chooses both the ports it serves and the subset of them it pilots by minimizing an expected free energy that trades achievable rate against information gain and port-switching cost. Selection is greedy and submodular, at $\mathcal{O}(NM)$ per slot. On a 441-port grid with a hybrid front end at $n_{\text{RF}} = 2K$, the agent attains 91% of a full-CSI genie’s sum rate while measuring only 2.3% of the candidate ports; pilots can then be withdrawn from ports that are still served, one fifth of them for 2% of the sum rate and two fifths for 8%. No training is required.

Index Terms—Fluid antenna systems, port selection, pilot allocation, active inference, expected free energy, partial CSI, Kalman filtering.

I. INTRODUCTION

Fluid antenna systems (FAS), also called movable- or reconfigurable-position antenna systems, replace fixed radiating elements with elements whose position can be adjusted among a dense set of candidate positions, or ports, within a compact aperture [1]. By moving to a favourable point of the spatial fading pattern, an FAS harvests diversity that a fixed array of the same size cannot, using only a handful of radio-frequency (RF) chains, since at any instant only the activated ports are connected to the front end. The pivotal control problem is port selection: deciding slot by slot which subset of the candidates to activate.

A port can be scored only if its channel is known, so conventional selection presupposes channel state information (CSI) at every candidate. That is expensive. The aperture is sampled at sub-wavelength spacing, so the ports are packed far more densely than a half-wavelength array and the number of coefficients to estimate grows rapidly with aperture. Worse, only activated ports can be observed at all. The bottleneck is therefore not the selection arithmetic but channel acquisition

under a sensing budget: the system must decide which ports are worth measuring — and, we add, whether a port that is served needs to be measured at all.

Two lines of work approach this and leave the gap open. Learning-based selection — deep reinforcement learning over ports [2] and online bandits [3] — learns policies directly, but treats the full channel as an observed, cost-free input and requires extensive training. Channel estimation for FAS [4], [5] exploits spatial correlation to reconstruct the channel from few pilots, but solves estimation in isolation: it recovers the channel without deciding what to sense or what to serve, and never weighs a measurement against its cost. What is missing is a formulation in which sensing and serving are chosen jointly under an explicit budget.

Active inference supplies that perspective. Rooted in the free-energy principle [6], it casts a decision-maker as an agent holding a generative model of its environment and acting to minimize an expected free energy (EFE), which decomposes into a pragmatic value rewarding goal attainment and an epistemic value rewarding informative actions. The fit to FAS is close: the hidden state is the channel at unmeasured ports, the generative model is its spatial and temporal correlation, the pragmatic value is throughput, and the epistemic value is what a pilot reveals about the ports left unseen. The exploit–explore tension that defines the problem is thus expressed by the EFE rather than engineered in by hand.

- 1) We formulate port selection *and pilot allocation* jointly as active inference under partial CSI. The piloted set is a strict subset of the activated set, making the pilot budget a decision variable rather than a fixed cost — to our knowledge the first such treatment for FAS.
- 2) We give a concrete agent: a Gaussian belief over all N ports from a spatial prior and an order- p autoregressive temporal model, with ports and pilots chosen by greedy EFE minimization at $\mathcal{O}(NM)$ per slot, and an exact reduced-rank belief that makes a 21×21 grid tractable.
- 3) We show that the agent reaches 91% of a full-CSI genie’s sum rate while measuring 2.3% of the ports, and degrades gracefully as pilots are further withdrawn from ports it continues to serve.

II. FLUID ANTENNA SYSTEM MODEL

We consider the downlink of a single-cell multiuser system in which a base station (BS) equipped with a fluid antenna serves K single-antenna users.

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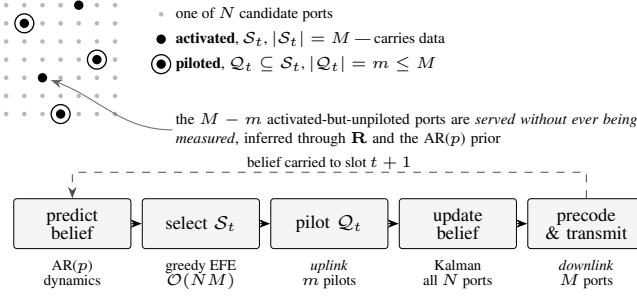


Fig. 1. Per-slot operation. The base station activates M of N candidate ports, but spends pilots on only $m \leq M$ of them; the remaining $M - m$ activated ports carry data while never being measured, their channels supplied by the belief alone. Both the activated set S_t and the piloted subset Q_t are chosen by minimizing the same expected free energy, which is what turns the pilot budget from a fixed cost into a design variable.

A. Array Geometry and Activation

The BS fluid antenna comprises $N = N_x \times N_y$ candidate ports on a uniform planar grid over a compact aperture with sub-half-wavelength spacing. It has M radio-frequency (RF) chains, so in slot t it activates a subset $S_t \subseteq \{1, \dots, N\}$ with $|S_t| = M$; only activated ports radiate and connect to the front end. Serving K streams requires $N \geq M \geq K$. For an index set $\mathcal{A}$, $\mathbf{P}_{\mathcal{A}} \in \{0, 1\}^{|\mathcal{A}| \times N}$ denotes the selection matrix extracting the entries of a vector on $\mathcal{A}$.

B. Spatio-Temporal Channel Model

Let $\mathbf{h}_k(t) \in \mathbb{C}^N$ be the channel from the N ports to user k in slot t . The ports being densely spaced, their coefficients are spatially correlated; under the classical Jakes model for rich scattering the correlation depends only on the separation d_{ij} ,

$$R_{ij} = J_0\left(\frac{2\pi d_{ij}}{\lambda}\right), \quad (1)$$

with $J_0(\cdot)$ the zeroth-order Bessel function of the first kind and λ the wavelength. Collecting these into $\mathbf{R}$, the small-scale fading is correlated Rayleigh, $\mathbf{h}_k \sim \mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$, with β_k the large-scale gain of user k . Sub-wavelength sampling makes $\mathbf{R}$ strongly rank-deficient, a fact Section III exploits.

The channel also ages between slots. Rather than the first-order Gauss–Markov model usual in this literature, we retain p past slots,

$$\mathbf{h}_k(t) = \sum_{i=1}^p a_i \mathbf{h}_k(t-i) + \mathbf{e}_k(t), \quad \mathbf{e}_k(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_v^2 \beta_k \mathbf{R}), \quad (2)$$

with $\mathbf{e}_k(t)$ independent of the past. The coefficients $\mathbf{a} = [a_1, \dots, a_p]^T$ and innovation variance σ_v^2 solve the Yule–Walker system $\mathbf{\Gamma} \mathbf{a} = \mathbf{r}$, $\sigma_v^2 = r(0) - \mathbf{a}^T \mathbf{r}$, for the Jakes temporal autocorrelation $r(\tau) = J_0(2\pi f_D T_s \tau)$, where $[\mathbf{\Gamma}]_{ij} = r(|i-j|)$, f_D is the maximum Doppler frequency and T_s the slot duration; $p = 1$ recovers the familiar first-order model.

C. Pilot Allocation Under Partial CSI

The BS cannot observe the full channel: a port yields information only when it is measured. Prior port-selection

work pilots every activated port, spending M pilot resources per slot. We separate the two decisions: in slot t the BS pilots a subset $Q_t \subseteq S_t$ with $|Q_t| = m \leq M$ of the ports it has activated, observing

$$\mathbf{y}_k(t) = \mathbf{P}_{Q_t} \mathbf{h}_k(t) + \mathbf{n}_k(t), \quad \mathbf{n}_k(t) \sim \mathcal{CN}(\mathbf{0}, \sigma_e^2 \mathbf{I}_m), \quad (3)$$

with σ_e^2 set by the pilot SNR. The unmeasured ports fall into two classes. Those outside S_t must be inferred if the BS is ever to select them. The $M - m$ ports in $S_t \setminus Q_t$ are more unusual: they carry data in this very slot while contributing no measurement, their channels supplied entirely by the belief through (1) and (2) (Fig. 1). Setting $m < M$ turns the pilot budget from a fixed overhead into a quantity the agent chooses.

D. Hybrid Transmission and Achievable Rate

The M activated ports are driven by $n_{\text{RF}} \leq M$ RF chains through a fully-connected network of unit-modulus phase shifters, giving the effective precoder

$$\mathbf{W} = \mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}} \in \mathbb{C}^{M \times K}, \quad |[\mathbf{F}_{\text{RF}}]_{ij}| = 1, \quad \|\mathbf{W}\|_F^2 \leq P. \quad (4)$$

Writing $\mathbf{h}_{k,S} = \mathbf{P}_S \mathbf{h}_k$ and treating inter-user interference as noise, user k achieves

$$R_k = \log_2 \left(1 + \frac{|\mathbf{h}_{k,S}^H \mathbf{w}_k|^2}{\sum_{j \neq k} |\mathbf{h}_{k,S}^H \mathbf{w}_j|^2 + \sigma^2} \right), \quad (5)$$

with σ^2 the receiver noise power. The target precoder is a regularized MMSE filter built from the belief rather than from measured CSI, so it accounts for residual uncertainty — dominant at the $M - m$ unpiloted ports — which $\mathbf{F}_{\text{RF}} \mathbf{W}_{\text{BB}}$ then factorizes by alternating minimization. For a fully-connected network with infinite-resolution shifters, $n_{\text{RF}} \geq 2K$ chains represent any digital precoder exactly [7], [8].

E. Switching Cost and Problem Formulation

Reconfiguring between slots costs delay and energy, charged through the number of ports that change, $C_t = |S_t \triangle S_{t-1}|$ with $\triangle$ the symmetric set difference. The design problem is

$$\begin{aligned} \max_{\{S_t, Q_t\}} \quad & \sum_{t=1}^T \left(\sum_{k=1}^K R_k(t) - \eta_{\text{sw}} C_t \right) \\ \text{s.t.} \quad & |S_t| = M, \quad Q_t \subseteq S_t, \quad |Q_t| = m, \end{aligned} \quad (6)$$

with $\eta_{\text{sw}} \geq 0$ trading throughput against reconfiguration cost. Its difficulty is that $R_k(t)$ depends on the channel at ports that (3) never reveals — the problem we address next through active inference.

III. ACTIVE INFERENCE FOR PORT SELECTION AND PILOT ALLOCATION

We now cast the problem as active inference. The agent holds a generative model of the channel, maintains a Bayesian belief over all N ports (perception), and each slot chooses the activated set and the pilot subset that minimize an expected free energy (action). The generative model needs no new ingredients: the prior is $\mathcal{CN}(\mathbf{0}, \beta_k \mathbf{R})$, the transition is the AR(p) dynamics (2), and the likelihood is the partial observation (3).

Preferences complete it, and here they are not a hand-tuned vector as in textbook active inference but the communication utility itself — the rate net of switching in (6) — so there is no arbitrary preference to set.

A. Perception: Complex Kalman Belief

Stacking the last p snapshots into $\mathbf{z}_k(t) = [\mathbf{h}_k(t)^T, \dots, \mathbf{h}_k(t-p+1)^T]^T \in \mathbb{C}^{pN}$ turns (2) into a first-order recursion $\mathbf{z}_k(t) = \mathbf{F}\mathbf{z}_k(t-1) + \boldsymbol{\epsilon}_k(t)$, with $\mathbf{F} = \mathbf{C}_a \otimes \mathbf{I}_N$ for $\mathbf{C}_a$ the companion matrix of $\mathbf{a}$, and process noise $(\mathbf{e}_1 \mathbf{e}_1^T) \otimes (\sigma_v^2 \beta_k \mathbf{R})$ confined to the current-time block. Observation (3) reads $\mathbf{y}_k(t) = \mathbf{G}\mathbf{z}_k(t) + \mathbf{n}_k(t)$ with $\mathbf{G} = [\mathbf{P}_{\mathcal{Q}_t} \ \mathbf{0}]$: pilots see the current channel only.

The model is linear-Gaussian, so the exact posterior is Gaussian, $q(\mathbf{z}_k) = \mathcal{N}(\hat{\mathbf{z}}_k, \mathbf{P}_k)$, maintained by a per-user complex Kalman filter. The predict step propagates the belief through the AR dynamics, encoding channel aging,

$$\hat{\mathbf{z}}_k \leftarrow \mathbf{F}\hat{\mathbf{z}}_k, \quad \mathbf{P}_k \leftarrow \mathbf{F}\mathbf{P}_k\mathbf{F}^H + \mathbf{Q}_k, \quad (7)$$

and the pilots on $\mathcal{Q}_t$ correct it through the Kalman gain $\mathbf{K}_k = \mathbf{P}_k \mathbf{G}^H (\mathbf{G}\mathbf{P}_k \mathbf{G}^H + \sigma_e^2 \mathbf{I}_m)^{-1}$,

$$\hat{\mathbf{z}}_k \leftarrow \hat{\mathbf{z}}_k + \mathbf{K}_k(\mathbf{y}_k - \mathbf{G}\hat{\mathbf{z}}_k), \quad (8)$$

$$\mathbf{P}_k \leftarrow (\mathbf{I} - \mathbf{K}_k \mathbf{G}) \mathbf{P}_k (\mathbf{I} - \mathbf{K}_k \mathbf{G})^H + \sigma_e^2 \mathbf{K}_k \mathbf{K}_k^H. \quad (9)$$

Only the $m \times m$ innovation covariance is inverted, regularized by $\sigma_e^2 \mathbf{I}_m$, so the rank-deficient $\mathbf{R}$ needs no artificial jitter, and the Joseph form (9) keeps $\mathbf{P}_k$ Hermitian positive semidefinite. The belief over the current channel is the leading block, $\boldsymbol{\mu}_k = \boldsymbol{\Pi} \hat{\mathbf{z}}_k$ and $\boldsymbol{\Sigma}_k = \boldsymbol{\Pi} \mathbf{P}_k \boldsymbol{\Pi}^H$ with $\boldsymbol{\Pi} = [\mathbf{I}_N \ \mathbf{0}]$. Two couplings then do the work, and together they are what permit $m < M$: spatially the off-diagonals of $\mathbf{R}$ carry a measurement to a port's neighbours, so m pilots inform far more than m ports, and temporally (7) carries information across slots, so a port piloted earlier retains a usable estimate now.

B. Exact Reduced-Rank Belief

Carrying $\mathbf{P}_k \in \mathbb{C}^{pN \times pN}$ is infeasible at our scale — $N = 441$, $p = 4$ give a 1764×1764 covariance per user per slot — and also unnecessary. Truncating $\mathbf{R} = \mathbf{B}\mathbf{A}\mathbf{B}^H$ to the r eigenvalues carrying all but 10^{-6} of the energy, and noting $\mathbf{h}_k(t) \in \text{range}(\mathbf{R})$ for every t , the factorization $\mathbf{h}_k(t) = \mathbf{B} \mathbf{c}_k(t)$ with $\mathbf{c}_k(t) \in \mathbb{C}^r$ holds *exactly*, so $\mathbf{c}_k$ is a sufficient statistic. The AR dynamics act identically on every port and carry over with $\mathbf{I}_N$ replaced by $\mathbf{I}_r$; the state falls from pN to pr and the per-slot cost from $\mathcal{O}((pN)^3)$ to $\mathcal{O}(pr^3)$. At $N = 441$ the numerical rank is $r = 26$: some 4.9×10^3 less arithmetic at no modelling cost.

C. Action: Expected Free Energy

The expected free energy of a policy is conventionally the sum of two terms: a pragmatic term, the expected log-preference over outcomes, and an epistemic term rewarding informative actions. Reconfiguration cost does not need a third term, because preferences are a modelling choice and ours are

simply that the agent prefers outcomes with high rate and few port changes,

$$\ln \tilde{p}(o_t | \mathcal{S}) = \sum_k R_k(t) - \eta_{\text{sw}} |\mathcal{S} \triangle \mathcal{S}_{t-1}| + \text{const}, \quad (10)$$

which is exactly the per-slot integrand of the system objective (6). The reconfiguration count is deterministic given the action, so its expectation is itself and the pragmatic value remains in closed form. Scoring a candidate pair $(\mathcal{S}, \mathcal{Q})$ therefore needs only the canonical two terms,

$$G(\mathcal{S}, \mathcal{Q}) = - \left[\text{Prag}(\mathcal{S}) - \eta_{\text{sw}} |\mathcal{S} \triangle \mathcal{S}_{t-1}| \right] - \beta_w \text{Epis}(\mathcal{Q}), \quad (11)$$

with $\beta_w > 0$ an exploration precision. The bracket is the expected log-preference (10) under the belief. Its first part is the sum rate predicted under the robust MMSE precoder, penalized by the belief's own residual uncertainty: with $\boldsymbol{\mu}_{k,S} = \mathbf{P}_S \boldsymbol{\mu}_k$, $\boldsymbol{\Sigma}_{k,S} = \mathbf{P}_S \boldsymbol{\Sigma}_k \mathbf{P}_S^H$,

$$\text{Prag}(\mathcal{S}) = \sum_k \log_2 \left(1 + \frac{|\boldsymbol{\mu}_{k,S}^H \mathbf{w}_k|^2}{\mathcal{I}_k} \right), \quad (12)$$

$$\mathcal{I}_k = \sum_{j \neq k} |\boldsymbol{\mu}_{k,S}^H \mathbf{w}_j|^2 + \sum_j \mathbf{w}_j^H \boldsymbol{\Sigma}_{k,S} \mathbf{w}_j + \sigma^2. \quad (13)$$

The middle term of (13) makes the agent conservative when uncertain: a port with large posterior variance contributes interference the precoder cannot null. It bites hardest at the $M - m$ ports served without being piloted, so (12) already prices the cost of not measuring them. The epistemic value is the mutual information the pilots on $\mathcal{Q}$ would yield,

$$\text{Epis}(\mathcal{Q}) = \sum_k \log_2 \det(\mathbf{I}_m + \sigma_e^{-2} \mathbf{P}_{\mathcal{Q}} \boldsymbol{\Sigma}_k \mathbf{P}_{\mathcal{Q}}^H). \quad (14)$$

Being a log-determinant rather than a sum of marginal variances, (14) prices redundancy: two candidates that are both uncertain *and* strongly correlated carry nearly the same information, so the second earns little once the first is measured.

D. Greedy Selection of Both Sets

Exact minimization of (11) is combinatorial in both arguments, so we build the sets greedily, each step adding the port whose inclusion most decreases G ,

$$p^* = \arg \max_{p \notin \mathcal{S}} \left[G(\mathcal{S}) - G(\mathcal{S} \cup \{p\}) \right], \quad (15)$$

until $|\mathcal{S}_t| = M$; the pilot subset is then built the same way, greedily maximizing (14) over $\mathcal{Q} \subseteq \mathcal{S}_t$ until $|\mathcal{Q}_t| = m$. Note that this is sequential greedy, not a ranking: because (14) is a log-determinant, the worth of a port depends on which ports are already in $\mathcal{Q}$. One objective governs both sets: eliminating $\mathcal{Q}$ by inner minimization gives a function of $\mathcal{S}$ alone, $\mathcal{S}_t = \arg \min_{\mathcal{S}} \min_{\mathcal{Q} \subseteq \mathcal{S}} G$, and choosing the two in sequence approximates that at $\mathcal{O}(NM)$ plus $\mathcal{O}(Mm)$ per slot. Since Epis is monotone and submodular and the switching term modular, greedy carries the standard $(1 - 1/e)$ guarantee on the information-driven part; in practice it is far tighter, and at this operating point $\binom{M}{m}$ is small enough to check against exhaustive search — greedy returns the exact optimum in 95%

of slots and never falls below 0.9996 of it. Order within the slot matters: we predict, select, pilot, update, and only then precode and transmit (Fig. 1), so that the piloted ports carry a posterior error of about σ_e^2 rather than the one-step aging floor a predict-then-transmit ordering would incur.

IV. NUMERICAL RESULTS

[TODO: Not yet written in the working draft. Figures exist in figures/.]

V. CONCLUSION

[TODO: Not yet written in the working draft.]

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